

Hospital Emergency Room Dashboard

1. Project Overview

The Hospital Emergency Room Dashboard is a data analytics project designed to analyze and visualize emergency room patient data. The project focuses on understanding important hospital emergency-room metrics such as patient volume, waiting time, admission status, patient demographics, department/referral information, and satisfaction levels.

The main objective is to transform raw hospital emergency-room data into meaningful insights using Microsoft Excel and Power BI. The dashboard helps healthcare management monitor patient flow, identify operational patterns, and make data-driven decisions to improve emergency-room efficiency and patient experience.

2. Dataset Summary

The project uses a **Hospital Emergency Room dataset** containing patient-level emergency-room records.

The dataset includes information related to:

Patient Information

- Patient ID
- Patient age
- Gender
- Patient demographics

Visit Information

- Admission/visit date
- Visit time
- Patient arrival information
- Department/referral information

Emergency Room Metrics

- Waiting time
- Patient satisfaction score
- Admission status
- Patient visit/encounter information

3. Tools and Technologies Used

The project was completed using the following tools:

Tool	Purpose
Microsoft Excel	Data loading, cleaning and initial analysis
PowerPoint / Gamma	Presentation of project findings
MS Word	Project documentation and final report

4. Exploratory Data Analysis

We began with data preparation and cleaning given below the dataset

Patient Id	Patient Admission Date	Patient First Initial	Patient Last Name	Patient Gender	Patient Age	Patient Race	Department Referral	Patient Admission Flag	Patient Satisfaction Score	Patient Waittime	Patient Admission Flag
145-39-5406	20-03-2024 08:47	H	Glasspool	M	69	White	None	FALSE	10	39	FALSE
316-34-3057	15-06-2024 11:29	X	Methuen	M	4	Native Ameri	None	TRUE		27	TRUE
897-46-3852	20-06-2024 09:13	P	Schubuser	F	56	African Amer	General Practice	TRUE	9	55	TRUE
358-31-9711	04-02-2024 22:34	U	Titcombe	F	24	Native Ameri	General Practice	TRUE	8	31	TRUE
289-26-0537	04-09-2024 17:48	Y	Gionettitti	Male	5	African Amer	Orthopedics	FALSE		10	FALSE
255-51-2877	20-04-2023 00:13	H	Buff	M	58	Asian	None	FALSE		59	FALSE
465-97-0990	23-08-2023 08:26	F	Perrat	F	68	White	None	TRUE		43	TRUE
157-31-7520	29-07-2023 16:57	K	Gwillim	F	47	Two or More	None	TRUE		23	TRUE
432-34-5614	19-02-2024 06:54	E	Dewhirst	F	79	White	None	FALSE	1	42	FALSE
609-17-8678	11-10-2024 05:25	M	Crebo	Male	62	African Amer	None	FALSE		51	FALSE
497-14-6812	26-07-2024 01:45	Q	Churchard	F	73	White	Gastroenterology	TRUE		34	TRUE
393-38-9502	10-03-2024 22:02	N	Corpes	F	16	White	Orthopedics	FALSE		39	FALSE
288-05-6370	12-11-2023 16:00	R	Brixey	F	16	Native Ameri	General Practice	TRUE		53	TRUE
784-54-9931	25-06-2023 09:40	M	Goudie	M	46	Pacific Island	None	FALSE		45	FALSE
662-21-6522	04-05-2023 13:16	G	Stanlack	Male	69	White	None	TRUE		49	TRUE
528-73-1801	19-09-2023 01:53	C	McMurty	M	37	Declined to I	None	TRUE		57	TRUE
370-19-2271	25-05-2024 22:11	I	Scothorn	F	50	Asian	General Practice	FALSE		35	FALSE
458-98-8860	25-06-2023 18:59	J	Helgass	M	37	Declined to I	None	FALSE		55	FALSE
728-31-2493	04-09-2023 16:15	V	Chittock	F	70	Asian	Physiotherapy	TRUE		50	TRUE
823-34-5523	16-11-2023 23:46	F	Prendergast	M	55	Asian	None	TRUE	2	40	TRUE
521-70-7472	30-06-2023 05:22	T	Bissiker	F	63	Native Ameri	None	TRUE		25	TRUE
344-36-7156	22-05-2023 16:48	M	Mandell	F	44	Asian	None	FALSE	2	51	FALSE
455-21-3671	17-11-2023 07:24	D	Coste	Female	11	Declined to I	None	FALSE	4	30	FALSE
259-10-0339	26-01-2024 05:58	Q	Dodridge	M	8	Asian	General Practice	FALSE		16	FALSE
720-54-2625	24-05-2023 14:42	C	Pavie	F	4	Native Ameri	None	TRUE		23	TRUE
561-92-7059	12-04-2023 21:02	Z	Sleightholm	M	42	African Amer	None	FALSE	0	51	FALSE
569-53-2833	01-09-2024 02:38	C	Moade	F	68	African Amer	None	FALSE		56	FALSE

The analysis focused on:

4.1 Total Patients

Calculated the total number of emergency-room patient records.

Daily trends of no. of patient	
Row Labels	Distinct Count of Patient Id
1-Feb	13
2-Feb	10
3-Feb	8
4-Feb	12
5-Feb	19
6-Feb	9
7-Feb	13
8-Feb	19
9-Feb	10
10-Feb	20
11-Feb	15
12-Feb	13
13-Feb	9
14-Feb	19
15-Feb	14
16-Feb	17
17-Feb	17
18-Feb	15
19-Feb	9
20-Feb	14
21-Feb	22
22-Feb	16
23-Feb	22
24-Feb	12
25-Feb	20
26-Feb	18

4.2 Average Waiting Time

Calculated the average time patients waited before receiving service.

average wait time	
Row Labels	Average of Patient Waittime
1-Feb	35.69
2-Feb	45.40
3-Feb	29.38
4-Feb	34.58
5-Feb	38.68
6-Feb	34.78
7-Feb	37.31
8-Feb	35.63
9-Feb	36.60
10-Feb	39.70
11-Feb	37.40
12-Feb	27.77
13-Feb	38.78
14-Feb	31.00
15-Feb	35.93
16-Feb	37.88
17-Feb	40.59
18-Feb	34.53
19-Feb	40.33
20-Feb	35.29
21-Feb	35.50
22-Feb	38.56
23-Feb	42.73
24-Feb	37.42
25-Feb	32.45
26-Feb	40.06

4.3 Patient Demographics

Analyzed patient distribution by:

- Gender
- Age
- Age groups

Gender wise analysis	
Row Labels	Count of Patient Gender
Female	194
Male	237
Grand Total	431.00

4.4 Admission Analysis

Compared:

- Admitted patients
- Non-admitted patients

Admission Status	No.of Patient	% Status	
Not Admitted	207	48%	
Admitted	224	52%	

4.5 Patient Satisfaction

Analyzed satisfaction scores to understand the patient experience.

Satisfaction score daily trend	
Row Labels	Average of Patient Satisfaction Score
1-Mar	7.20
2-Mar	6.00
3-Mar	1.50
4-Mar	2.75
5-Mar	5.00
6-Mar	5.50
7-Mar	5.09
8-Mar	7.67
9-Mar	3.50
10-Mar	3.67
11-Mar	3.80
12-Mar	1.00
13-Mar	7.00
14-Mar	5.00
15-Mar	4.25
16-Mar	4.00
17-Mar	7.33
18-Mar	9.00
19-Mar	5.25
20-Mar	6.60
21-Mar	6.25
22-Mar	6.33
23-Mar	7.00
24-Mar	5.67
25-Mar	3.33
26-Mar	4.75

4.6 Visit Trends

Analyzed emergency-room visits based on:

- Date
- Month
- Day
- Time/period

attended status	
Row Labels	Count of Patient attend Status
Delay	312
On time	194
Grand Total	506.00

5. Data Analysis using Excel

Excel Analysis

Excel was used to perform calculations and create supporting analysis.

Important calculations can include:

5.1 Total Patients

COUNT / COUNTA of Patient ID

5.2 Average Waiting Time

AVERAGE(Waiting Time)

5.3 Average Satisfaction

AVERAGE(Satisfaction Score)

5.4 Admission Rate

Admitted Patients / Total Patients × 100

5.5 Patient Distribution

Patients can be grouped according to:

- Gender
- Age Group
- Admission Status
- Referral Department
- Visit Date

6. Dashboard Visualizations

Recommended visuals include:

6.1 Patients by Gender

A donut or column chart showing male vs. female patient distribution.

6.2 Patients by Age Group

A bar chart showing patient volume across different age groups.

6.3 Patients by Admission Status

A donut chart comparing admitted and non-admitted patients.

6.4 Average Waiting Time

A KPI/card or gauge showing average waiting time.

6.5 Patient Visits Over Time

A line chart showing patient volume by date/month.

6.6 Patients by Department/Referral

A bar chart showing the number of patients associated with each department or referral area.

6.7 Patient Satisfaction

A chart showing the distribution or average satisfaction score.

7. Dashboard Interactivity

Possible slicers include:

- Date
- Gender
- Age Group
- Admission Status
- Department/Referral
- Patient Type

For example, selecting a particular **month** can update the KPI cards and charts to show only emergency-room activity during that period.

8. Business Recommendations

Based on the analysis, the following recommendations can be considered:

8.1 Reduce Patient Waiting Time

Hospital management can analyze peak periods and allocate additional staff during high-volume periods to reduce waiting times.

8.2 Improve Staff Allocation

Staffing levels can be adjusted according to patient-volume trends identified in the dashboard.

8.3 Monitor Peak Hours

Emergency-room management should closely monitor periods with unusually high patient volumes and prepare resources accordingly.

8.4 Improve Patient Experience

Waiting time and satisfaction should be monitored together to identify areas where service improvements can increase patient satisfaction.

8.5 Monitor Admission Trends

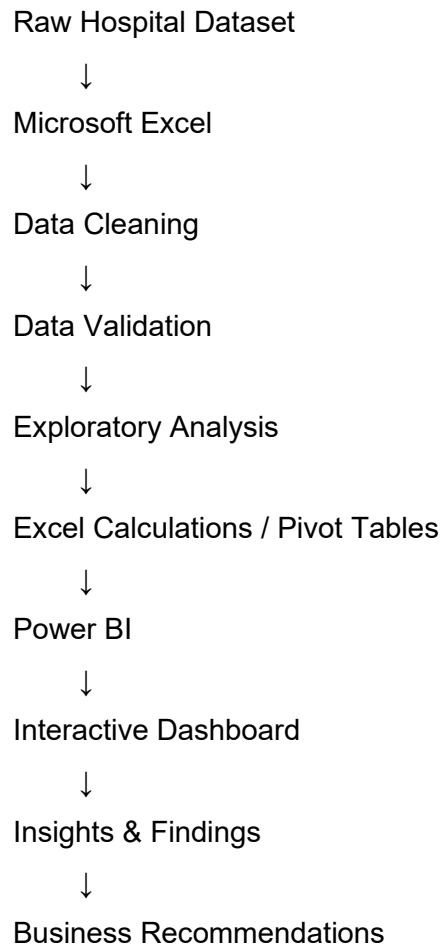
Admission patterns can help management understand emergency-room workload and plan beds and other hospital resources.

8.6 Use Data for Continuous Monitoring

The Power BI dashboard can be updated regularly so that hospital management can monitor important KPIs and identify changes in emergency-room performance.

9. Project Workflow

You can include this simple workflow diagram in your report:

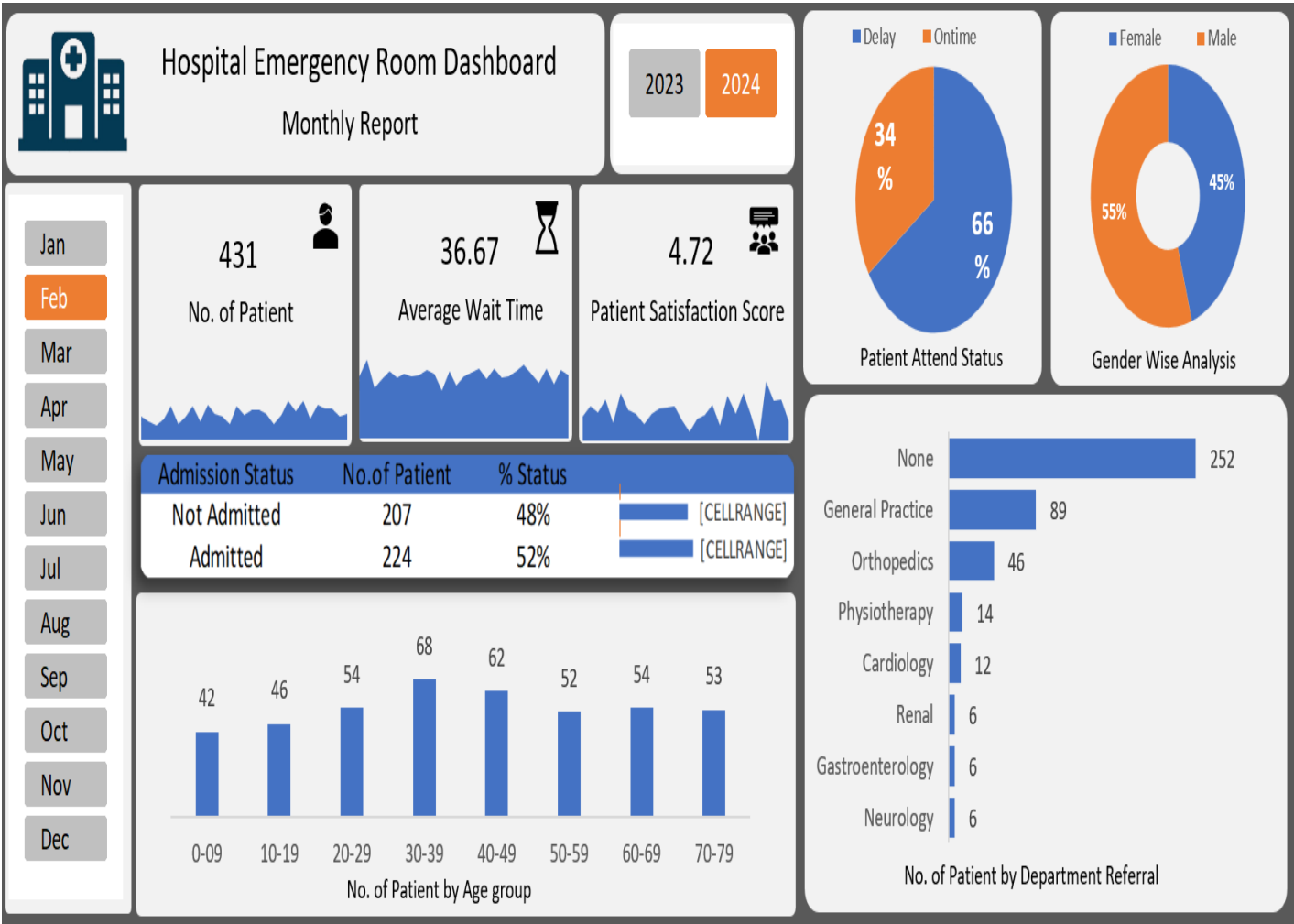


10. Project Outcome

The **Hospital Emergency Room Dashboard** successfully transforms raw emergency-room data into an interactive analytical solution.

The project provides a clear view of:

- Patient volume
- Waiting time
- Admission patterns
- Patient demographics
- Patient satisfaction
- Visit trends
- Department/referral patterns



The dashboard enables healthcare stakeholders to quickly understand emergency-room performance and supports **data-driven operational decision-making**.

11.Conclusion

- The Hospital Emergency Room Dashboard demonstrates how data analytics tools can be used to convert healthcare data into meaningful business insights.
- **Microsoft Excel** was used for data preparation, cleaning, calculations, and preliminary analysis, while it used to develop an interactive dashboard.
- The project provides a practical framework for monitoring emergency-room activity.
- Identifying opportunities to improve patient flow, waiting time, resource allocation, and patient experience.